

Statistics Homework 2: Effect Size and Error

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Part A: Definitions

Match each definition with the correct term.

1. The probability of committing a type I error if the null hypothesis is true.
2. The probability of committing a type I error if the null hypothesis is false.
3. A word that means: "if the null hypothesis is true, then the results from the sample are extremely unusual."
4. The probability of making a correct decision if the null hypothesis is false.
5. The probability of committing a type II error if the null hypothesis is false.
6. A person's average score on a test if he or she were assessed an infinite number of times.
7. The assumption that the population parameter for a statistic is equal to zero.
8. A distribution of sampling error that is almost normal in shape.
9. A distribution of sampling error that is skewed in shape.
10. An effect size that is typically used when one variable is dichotomous.
11. An effect size that is typically used with two numerically scales variables.

A. Power
B. Significant
C. Alpha
D. d
E. r
F. f

G. Null hypothesis
H. Beta
I. Zero
J. True score
K. t

Part B: Distributions

Use the key to answer the next 6 questions.

12. Standard deviation pertains to:
13. Standard error of measure pertains to:
14. Standard error of the mean difference pertains to:
15. Standard error of estimate pertains to:
16. Standard error of the correlation pertains to:
17. Standard error of the mean pertains to:

Key	A. Observations of different assessments from a case B. Observations of different cases from a sample C. Observations of different samples from a population
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Use the key to answer the next 6 questions.

18. Standard deviation reflects:
19. Standard error of measure reflects:
20. Standard error of the mean difference reflects:
21. Standard error of estimate reflects:
22. Standard error of the correlation reflects:
23. Standard error of the mean reflects:

Key	A. Deviations around a true score B. Deviations around a sample mean C. Deviations around a predicted value D. Deviations around a population parameter
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Part C: Identifying terms and formulas

Use the following key to answer the next 8 questions.

Key	A	B	C	D	E
	$\frac{sd_{true}^2}{sd^2}$	$sd^2 - sd_{Measure}^2$	$sd\sqrt{1 - \text{reliability}}$	$sd_{\bar{x}}$	sd_{est}

24. This is estimated by a correlation between two alternate forms.
25. This statistic describes a distribution that becomes almost perfectly normal in shape as the sample size increases beyond 30.
26. This is equal to true variance.
27. This is estimated by a test retest correlation.
28. This will become smaller as the sample size increases.
29. This is the standard error of measure.
30. This value will become smaller as the effect size increases.
31. This is estimated by Cronbach's alpha.

Part D: Soft and flat emotions

Soft emotion and flat emotion are two types of emotion that people sometimes experience during relationship conflicts. Is there a gender difference in the extent to which people report experiencing these two emotions? Sanford (2007) investigated soft and flat emotion in a sample of 700 married women and 524 married men. Each participant was asked to think of a recent episode of marital conflict and to answer questions regarding his or her emotions at the time of the conflict. Soft emotion was measured by a 4-item scale measuring the extent to which a person felt "hurt," "sad," "concerned," or "disappointed." Cronbach's alpha was used to estimate reliability. Alphas were .78 for wives and .75 for husbands. Flat emotion was measured by a 4-item scale measuring the extent to which a person felt "bored," "uninterested," "indifferent," and "disengaged." Alphas were .81 for wives and .85 for husbands.

The results from this study are listed below. Use these results to compute the effect size (d-statistic) for the difference between women and men on both soft emotion and flat emotion. Also answer questions regarding the reliability of the variables.

Descriptive Statistics

	sex	N	Minimum	Maximum	Mean	Std. Deviation
soft	Men	524	.00	4.00	2.13	.94
	Women	700	.00	4.00	2.61	.98
flat	Men	524	.00	3.75	.64	.71
	Women	700	.00	3.50	.51	.65
Valid N (listwise)	Men	524				
	Women	700				

For soft emotion

32. What is the sum of squares for men?
33. What is the sum of squares for women?
34. What is the total degrees of freedom ($df_1 + df_2$)?
35. What is the pooled standard deviation?
36. What is the d-statistic?

37. Which of the following best describes the effect for soft emotion?

- A. trivial B. small C. medium D. large

For flat emotion

38. What is the sum of squares for men?

39. What is the sum of squares for women?

40. What is the total degrees of freedom ($df_1 + df_2$)?

41. What is the pooled standard deviation?

42. What is the d-statistic?

43. Which of the following best describes the effect for flat emotion?

- A. trivial B. small C. medium D. large

Reliability questions

Hint: To answer these questions, you will need to look at the results reported above regarding Crombach's alpha.

44. Which of the following best describes the reliability of **wives' soft emotion**?

- A. not adequate B. adequate C. good

45. For **wives' soft emotion**, what percent of the total variance is true variance?

46. For **wives' soft emotion**, what percent of the total variance is measurement error variance?

47. If an individual **wife** was assessed on repeated occasions, how much would each of her observed scores for **soft emotion** typically deviate from her true score?

48. Which of the following best describes the reliability of **husbands' soft emotion**?

- A. not adequate B. adequate C. good

49. For **husbands' soft emotion**, what percent of the total variance is true variance?

50. For **husbands' soft emotion**, what percent of the total variance is measurement error variance?

51. If an individual **husband** was assessed on repeated occasions, how much would each of his observed scores for **soft emotion** typically deviate from his true score?

52. Which of the following best describes the reliability of **wives' flat emotion**?

- A. not adequate B. adequate C. good

53. For **wives' flat emotion**, what percent of the total variance is true variance?

54. For **wives' flat emotion**, what percent of the total variance is measurement error variance?

55. If an individual **wife** was assessed on repeated occasions, how much would each of her observed scores for **flat emotion** typically deviate from her true score?

56. Which of the following best describes the reliability of **husbands' flat emotion**?

- A. not adequate B. adequate C. good

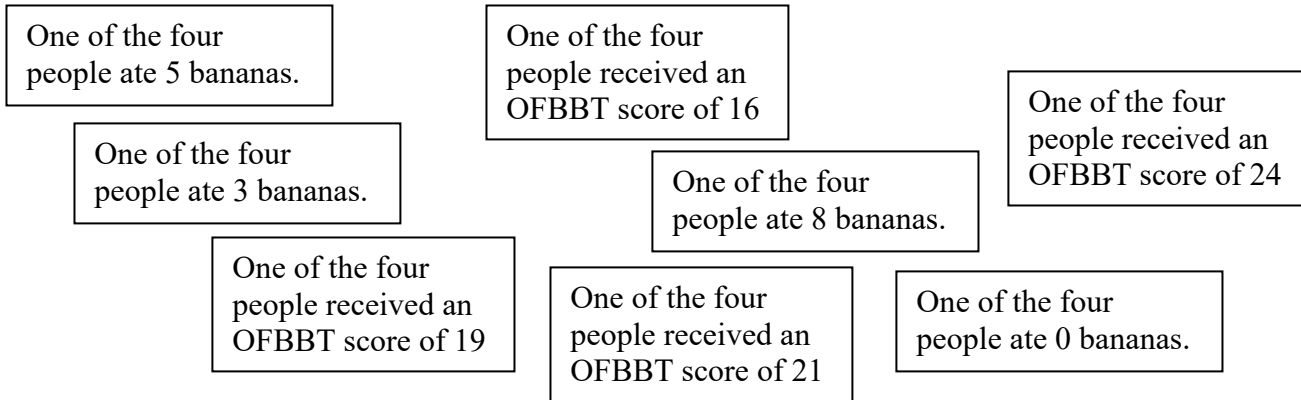
57. For **husbands' flat emotion**, what percent of the total variance is true variance?

58. For **husbands' flat emotion**, what percent of the total variance is measurement error variance?

59. If an individual **husband** was assessed on repeated occasions, how much would each of his observed scores for **flat emotion** typically deviate from his true score?

Part E: Covariance story problem

The following data are fictitious. Professor Dan Bana dressed up in a gorilla costume and went downtown to a public park and asked people to participate in a study. Each participant was asked to take the One Foot Banana Balancing Test (OFBBT). This test involves standing on one foot while peeling a banana and singing the happy banana song. After taking this test, each participant was given an opportunity to eat as many bananas as he or she wanted. Professor Bana hypothesized that scores on the OFBBT would correlate with the number of bananas eaten. Professor Dan Bana collected data from four people. The scores are listed below.

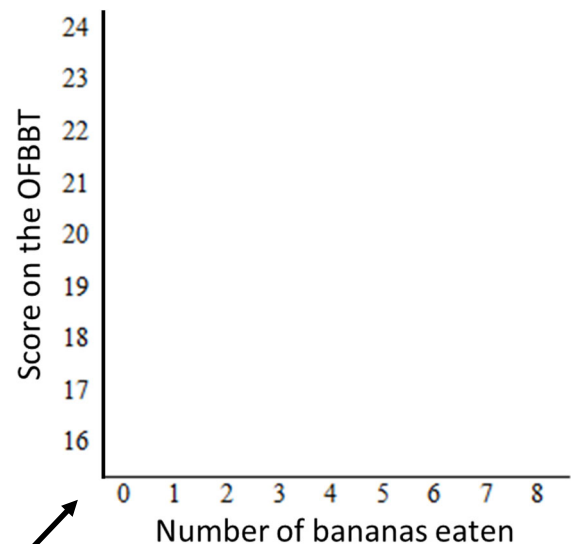


The information above lists all that data that were obtained in this study. However, it is not listed in any particular order, and the scores are not paired together. Although it is clear that each person in the study received one score on the OFBBT and a second score for the number of bananas eaten, the information above does not clarify which pairs scores belong to the same person. For the questions below, you will need to experiment with different ways that these scores could be paired together.

Given the scores listed above, how would you pair these scores together so that the dataset produces the **highest possible covariance** that this particular sample could produce?

Decide how to pair the scores together to produce the highest possible covariance, and then fill in the table below by entering the pairs of scores you selected for each person. Then plot the data for the four participants on the scatter plot. To do this, you will place four dots on the scatter plot, one for each participant.

	Number of bananas eaten	Score on the OFBBT
Person 1		
Person 2		
Person 3		
Person 4		



Enter your selected pairs of scores on this table.

Plot your selected pairs of scores on this scatter plot. For example, if you paired scores so that the person who ate 5 bananas scored 19 on the OFBT, then place a dot on the scatter plot that is directly above the number 5 on the banana scale and directly to the right of the number 19 on the OFBT scale.

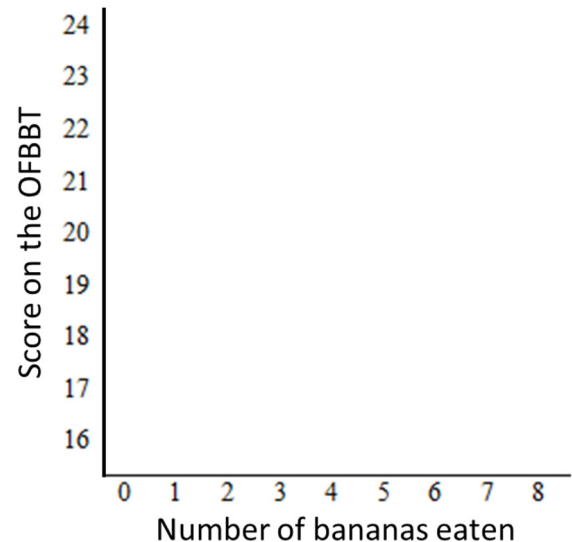
60. You just paired scores together so that the dataset would produce the highest possible covariance that this particular sample could produce. When you plotted your scores on the scatterplot, what shape was formed by the four points?
- Straight line
 - Square
 - Some other shape
61. You just paired scores together so that the dataset would produce the highest possible covariance that this particular sample could produce. Calculate that covariance and enter your answer on Canvas.

Negative covariance:

Now, given the scores listed on the previous page, how would you pair these scores together so that the dataset produces the **most negative possible covariance** that this particular sample could produce? In other words, how would you pair the scores together so that the covariance is a negative number and so that it is the lowest possible negative number it can be?

Decide how to pair the scores together to produce the lowest possible covariance, and then fill in the table below by entering the pairs of scores you selected for each person. Then, plot the data for the four participants on the scatter plot.

	Number of bananas eaten	Score on the OFBBT
Person 1		
Person 2		
Person 3		
Person 4		



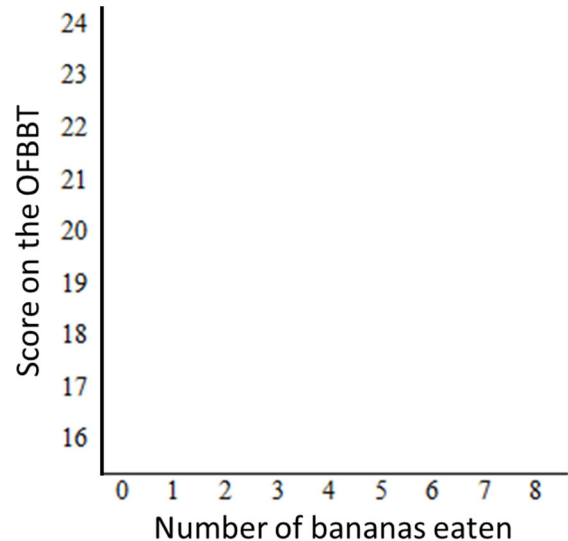
62. When you plotted your scores on the scatterplot, what shape was formed by the four points?
- Straight line
 - Square
 - Some other shape
63. You just paired scores together so that the dataset would produce the lowest (most negative) possible covariance that this particular sample could produce. Calculate that covariance and enter your answer on Canvas.

Zero association:

Now, given the scores listed on the previous page, how would you pair these scores together so that the dataset produces **a covariance that is zero or as close to zero as possible**? In other words, how would you pair the scores together to produce the lowest possible absolute value for the covariance?

Decide how to pair the scores together to produce a covariance of zero or as close to zero as possible, and then fill in the table on the next page by entering the pairs of scores you selected for each person. Then, plot the data for the four participants on the scatter plot.

	Number of bananas eaten	Score on the OFBBT
Person 1		
Person 2		
Person 3		
Person 4		



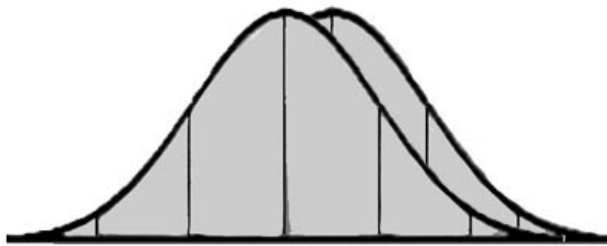
64. When you plotted your scores on the scatterplot, what shape was formed by the four points?

- D. Straight line
- E. Square
- F. Some other shape

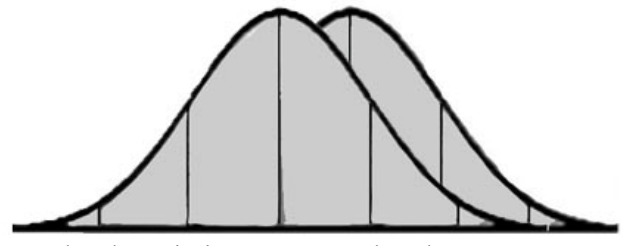
65. You just paired scores together to produce a covariance of zero or as close to zero as possible. Calculate that covariance and enter your answer on Canvas.

Part F: Recognizing effect sizes

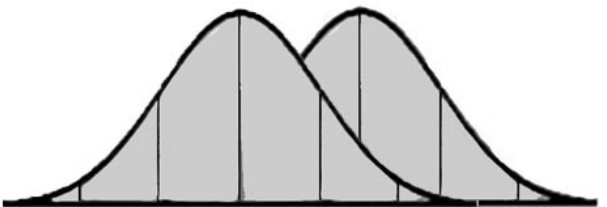
You should be able to provide a good estimate of the size of a d-statistic just by look at a figure showing two normal curves. Look at the figures below and provide a rough approximation of the d-statistic.



66. The d-statistic appears to be about:
(A) .1 (B) .5 (C) .8 (D) 1.0 (E) 2.0



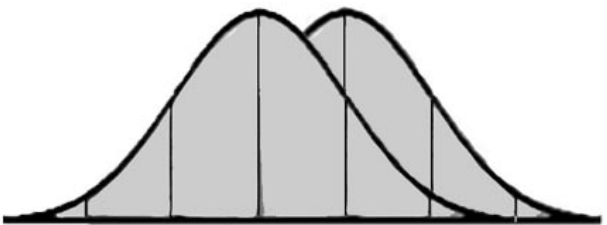
67. The d-statistic appears to be about:
(A) .2 (B) .5 (C) .8 (D) 1.5 (E) 5.0



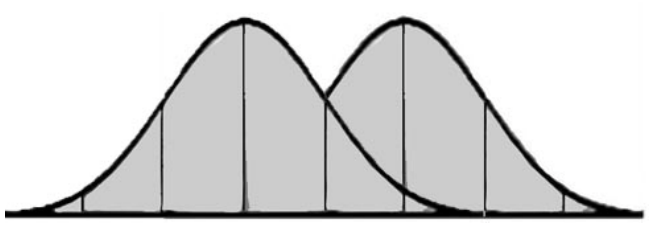
68. The d-statistic appears to be about:
(A) .3 (B) .8 (C) 1.2 (D) 1.5 (E) 2.0



69. The d-statistic appears to be about:
(A) .2 (B) .5 (C) .8 (D) 1.0 (E) 2.0



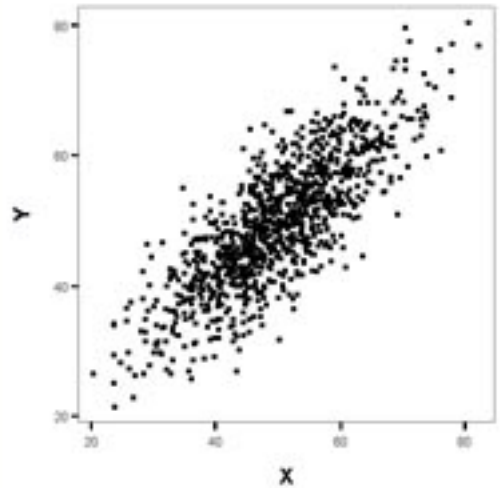
70. The d-statistic appears to be about:
(A) .1 (B) .5 (C) 1.0 (D) 1.5 (E) 2.0



71. The d-statistic appears to be about:
(A) .2 (B) .8 (C) 1.5 (D) 2.0 (E) 3.0

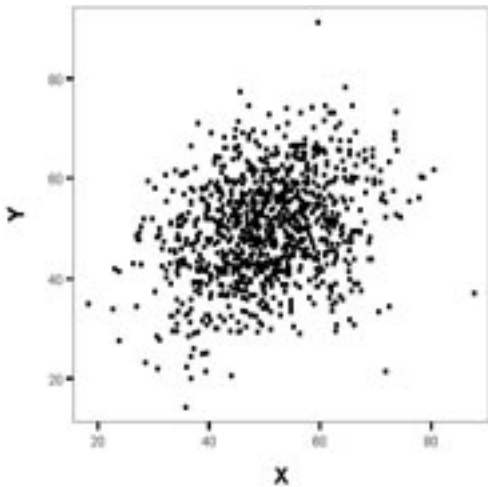
You should be able to provide a good estimate of the size of a correlation just by look at a scatter plot. Look at the following figures and provide a rough approximation of the correlations.

72. The correlation appears to be about:



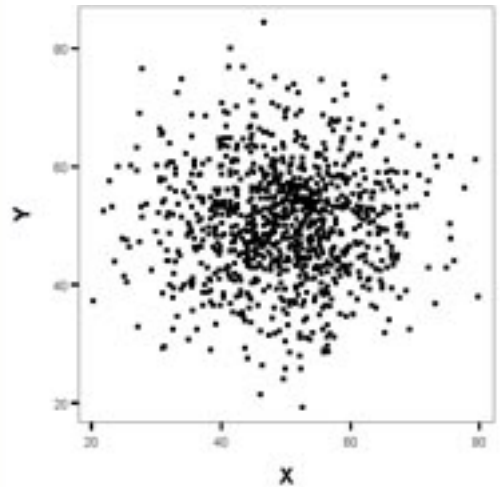
- A 0
- B .1
- C .3
- D .5
- E .8

73. The correlation appears to be about:



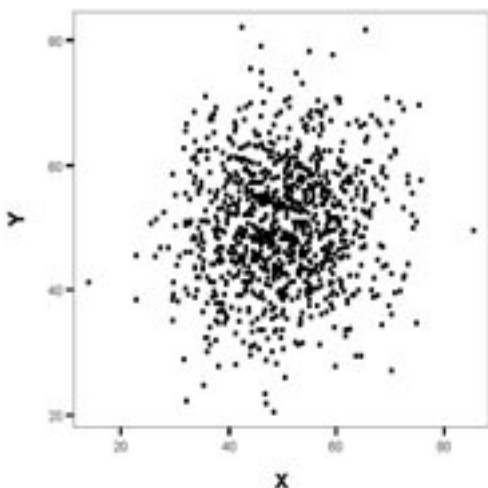
- A 0
- B .3
- C .6
- D .8
- E 1.0

74. The correlation appears to be about:



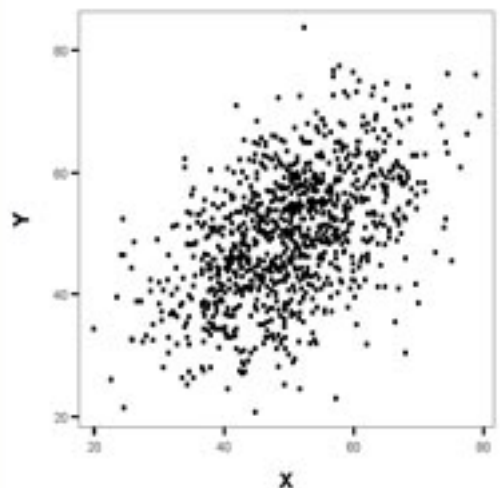
- A 0
- B .4
- C .6
- D .8
- E 1.0

75. The correlation appears to be about:



- A .1
- B .4
- C .6
- D .8
- E 1.0

76. The correlation appears to be about:



- A 0
- B .2
- C .5
- D .8
- E 1.0

Part G: More covariance questions

Assume that the variance of X is 18 and the variance of Y is 18:

77. If the correlation between X and Y is 1, what is the covariance?
78. If the correlation between X and Y is .5, what is the covariance?
79. If the correlation between X and Y is 0, what is the covariance?
80. Assume that the variance of X is 18 and the variance of Y is 18, and the correlation between X and Y is .5. If both X and Y are transformed into z-scores, what is the covariance of the z-scores?

Study tip:

The answers to the questions in this section should be immediately obvious. If you think you need a calculator to answer these questions, you are missing the obvious answer.

Part H: Computing a correlation

Use the hypothetical dataset below and compute the correlation between the two variables.

	Score on variable 1	Score on variable 2
Person 1	7	37
Person 2	12	58
Person 3	13	52
Person 4	11	46
Person 5	11	30
Person 6	6	47

Computing the standard deviation for variable 1

81. What is the sum of squares for variable 1?
82. What is the variance for variable 1?
83. What is the standard deviation for variable 1?

Computing the standard deviation for variable 2

84. What is the sum of squares for variable 2?
85. What is the variance for variable 2?
86. What is the standard deviation for variable 2?

Computing the covariance

87. What is the sum of cross products?
88. What is the covariance?

Computing the correlation

89. What is the correlation?
90. Which of the following best describes the size of this correlation?
A. trivial B. small C. medium D. large

Part I: questions about the t -statistic

A researcher analyzes data for a study and obtains the following statistics: $\bar{x} = 12$, and $sd_{\bar{x}} = 3$

Answer the following questions about calculating a t -statistic for the data in this example.

91. What is the t -value?
92. In the t -statistic formula, what is the assumed value for the population parameter?
93. Given the assumed value for the population parameter, what is the value that indicates the amount of sampling error that was made in this study?
94. If this study were repeated over and over, what is the typical amount of sampling error that would be made in each study?
95. Given the assumed value for the population parameter, the error in this study is equal to the typical error times what value?
96. One component of the t -formula is something that is calculated from the sample, and if we knew the true population parameter for this, there would be no reason for doing the t -test. In the above example, what is this value?
97. Another component of the t -formula is something that is calculated from the sample, and if we knew the true population parameter for this, the distribution would change from a t -distribution to a z -distribution. In the above example, what is this value?

Study tip:

You should be able to answer all the questions in this section regardless of whether you are given: (a) a mean and standard error of the mean, (b) a difference between means and standard error of difference, (c) a correlation and standard error of correlation, or (d) a beta and standard error of beta.

Part J: Linear equations

Fill in the blanks using the appropriate linear equations (or pieces of the linear equations).

Use the following key to answer each question. When entering your answers, simply type the letter of the correct term.

A. $x_{ai} = \text{true}_i + e_{ai}$	H. e_{sp}
B. $(\bar{y} - \bar{y})_{sp} = (\mu - \mu)_p + e_{sp}$	I. e_{ai}
C. $\bar{x}_{sp} = \mu_p + e_{sp}$	J. e_{is}
D. $y_{is} = (\text{prediction}_s \text{ based on } x_{is}) + e_{is}$	K. i and s stand for individual and sample
E. $x_{is} = \bar{x}_s + e_{is}$	L. a and i stand for assessment occasion and individual
F. $\beta_{sp} = \beta_p + e_{sp}$	M. s and p stand for sample and population
G. $r_{sp} = \rho_p + e_{sp}$	

A **standard deviation** indicates the typical magnitude of [98]_____ from the linear equation [99]_____, where the subscripts [100]_____.

A **standard error of measure** indicates the typical magnitude of [101]_____ from the linear equation [102]_____, where the subscripts [103]_____.

A **standard error of a mean difference** indicates the typical magnitude of [104]_____ from the linear equation [105]_____, where the subscripts [106]_____.

A **standard error of estimate** indicates the typical magnitude of [107]_____ from the linear equation [108]_____, where the subscripts [109]_____.

A **standard error of the correlation** indicates the typical magnitude of [110]_____ from the linear equation [111]_____, where the subscripts [112]_____.

A **standard error of a mean** indicates the typical magnitude of [113]_____ from the linear equation [114]_____, where the subscripts [115]_____.

Part K: Effect size examples

For each example listed below, provide the name of the statistic used to indicate effect size and the value of the effect size (based on the meta-analysis results discussed in class).

The effectiveness of cognitive therapy for treating depression

116. What is the name of the statistic?

117. What is the value of the effect size?

The association between therapeutic alliance and psychotherapy outcome

118. What is the name of the statistic?

119. What is the value of the effect size?

The association between clinician confidence and accuracy

120. What is the name of the statistic?

121. What is the value of the effect size?

The effectiveness of dialectical behavior therapy for treating borderline personality disorder

122. What is the name of the statistic?

123. What is the value of the effect size?

The association between self-compassion and (lack of) psychopathology

124. What is the name of the statistic?

125. What is the value of the effect size?

The effectiveness of prolonged exposure for treating posttraumatic stress disorder

126. What is the name of the statistic?

127. What is the value of the effect size?

The effectiveness of cognitive behavior therapy with motivational interviewing for treating alcohol use disorder

128. What is the name of the statistic?

129. What is the value of the effect size?

The association between marriage quality and life well-being

130. What is the name of the statistic?

131. What is the value of the effect size?

The effectiveness of behavioral couple therapy for treating relationship distress

132. What is the name of the statistic?

133. What is the value of the effect size?

The association between parenting and child delinquency

134. What is the name of the statistic?

135. What is the value of the effect size?

Part L: Written summaries

The “Flourishing Scale” is a questionnaire that measures the extent to which people feel like they are flourishing in life. Respondents use a 7-point scale to rate 8 different questions about life satisfaction and life quality. A total score is created by summing all 8 questions. Based on results from a recent study, Cronhbach’s alpha for the Flourishing Scale is .83, and descriptive statistics for this scale are printed below.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
FS_8	482	8.00	56.00	43.2220	9.41084
Valid N (listwise)	482				

136. Make the necessary calculations and **write a summary of the reliability of the Flourishing scale**. Your written summary should follow the template provided in your notes and described in the lecture add-on video. After you have provided a written summary, use the scoring system in the box below to grade your own answer. On Canvas, you will be asked to report if you wrote this summary and graded your own answer. The scoring system in the box below is the same as the scoring system that will be used to grade written summaries on learning assessments.

Scoring for a written summary of the reliability of a scale

This question is worth 5 points. Give yourself 1 point for correctly including each of the following components in your summary.

1. The reliability needs to be correctly labeled as either poor, adequate, or good. In this case, it should be labeled “good.”
2. The summary needs to report: Cronbach’s Alpha = .83
3. The summary needs to state the percent of variance that is measurement error. In this case, 17% of the variance is measurement error.
4. The summary needs to make it clear that you understand the meaning of the standard error of measure. In this case, the standard error of measure is 3.88; so this means that, on average, a person’s observed score is likely to deviate from her or his true score by 3.88 units. This definition needs to include the word “observed” (or a phrase with the same meaning) and the word “true” (or a phrase with the same meaning) and use wording consistent with the fact that each true score pertains to a single person (and not to a group of people, sample, or population).
5. The summary needs to use the correct value for the standard error of measure. In this case, you should have determined that the value is 3.88.

Deduct one-half point for problems in any of the following areas:

- To make written summaries easy to read, values should be rounded to two decimal places.
- The summary needs to name the scale being summarized. In this case, the scale should be called the “Flourishing Scale.”

Below is an example of an answer that would receive full credit:

The Flourishing Scale had good reliability (Cronbach’s Alpha = .83). These results suggest that 17% of the variance was measurement error and that on average, a participant’s observed score is likely to deviate from her or his true score by 3.88 units.

References

- Sanford, K. (2007). The Couples Emotion Rating Form: Psychometric properties and theoretical associations. *Psychological Assessment, 19*, 411-421.